

Project Design Phase

Problem – Solution Fit Template

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Project Name: Plugging into the Future: An Exploration of Electricity Consumption Patterns

Target Customers

- Government Energy Departments
- Electricity Distribution Companies
- Energy Analysts
- Researchers and Students
- Policy Makers
- Utility Managers

Customer Problems

- Electricity data is scattered across reports and spreadsheets.
- Difficult to compare state-wise and region-wise consumption.
- Traditional reports lack interactive visualizations.
- Manual analysis is time-consuming.
- Limited support for quick decision-making.

Existing Alternatives

- Excel spreadsheets
- Government PDF reports
- Static charts and tables
- Manual data analysis

Problems with Existing Alternatives

- No interactive filters.
- Time-consuming analysis.
- Limited geographical visualization.
- Difficult to compare multiple years.

Proposed Solution

- Interactive Tableau dashboard embedded in a Flask web application.
- India map visualization.
- State-wise and region-wise analysis.
- Year-wise comparison (2019 & 2020).
- Interactive filters, pie charts, and bar charts.

Customer Benefits

- Faster analysis.
- Easy comparison of electricity consumption.
- Improved decision-making.
- Better visualization of trends.
- Accessible through a web interface.

Value Proposition

An interactive electricity consumption analytics platform that transforms raw energy data into meaningful visual insights, enabling faster analysis, improved planning, and informed decision-making.

Problem – Solution Fit Summary

Problem	Solution	Customer Benefit
Scattered data	Centralized Tableau dashboard	Easy access
Difficult comparison	Interactive maps & filters	Quick analysis
Static reports	Dynamic charts	Better insights
Manual analysis	Automated visualizations	Saves time
Limited insights	Year & region analysis	Better decisions

Expected Outcome

The proposed solution helps users analyze electricity consumption efficiently through interactive dashboards, enabling better planning, trend identification, and data-driven decisions.